

What they forgot to teach you about R

Scientific workflows: Tools and Tips 

2023-04-20

Who am I?

- Ecologist, PhD student for some years and now scientific programmer

What is this lecture series?

Scientific workflows: Tools and Tips 

. . .



Every 3rd Thursday



4-5 p.m.



Webex

What they forgot to teach you about R

It's a book by J. Bryan and J. Hesters



Artwork by Allison Horst, CC BY 4.0

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... can make even small changes frustrating and difficult.

A cartoon illustration of a red, round character wearing a white chef's hat, looking distressed and sweating. A speech bubble above the character says "oh just add SALT!?!". In the background, there is a kitchen scene with a stove, pots, and a blender. A clock in the top left corner shows the time as 1:50 and has the text "BEEP BEEP BEEP!!" above it. The character is holding a book titled "RECIPE".

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Background

- Reproducibility 
 - Can someone else reproduce my results?
- Reliability 
 - Will my code work in the future?
- Reusability 
 - Can someone else actually use my code?

Today: Talk **best practice** rules to write clean, clear and maintainable code.

In other words: How to clean the kitchen?

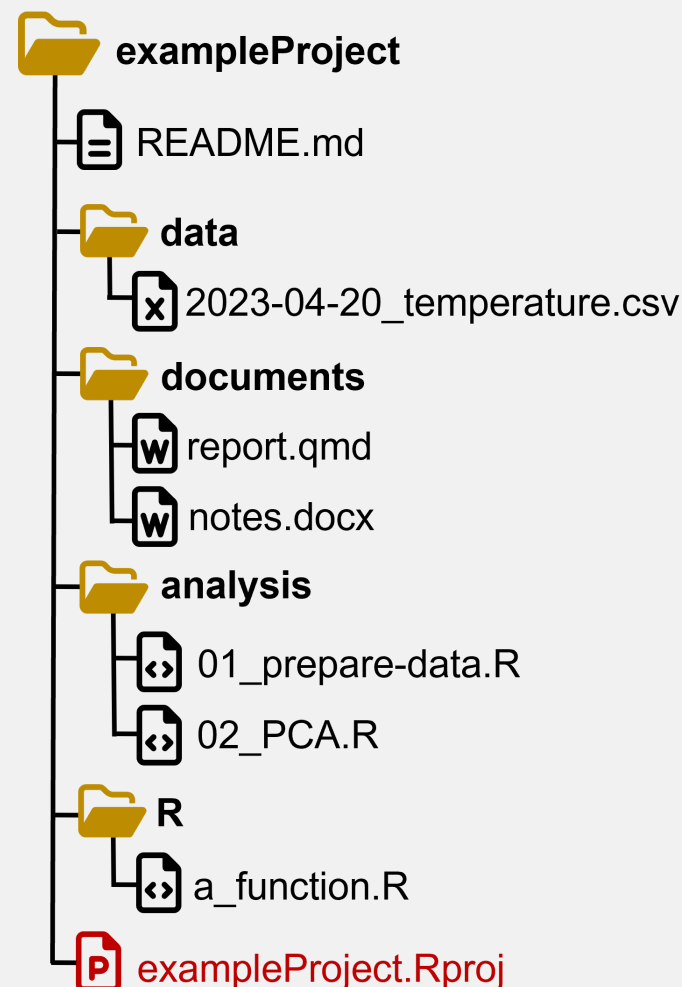
First things first

Project setup and structure

Use R Studio projects

Always make your project an R Studio Project (if possible)!

- Keep your files together
- An R Studio Project is just a normal directory with an `*.Rproj` file
 - double-click this file to open your project in R Studio
- Advantages:
 - Easy to navigate in R Studio
 - Project root is the working directory
 - Open multiple projects in separate R Studio instances



Create an R Studio Project

From scratch:

1. `File -> New Project -> New Directory -> New Project`
2. Enter a directory name (this will be the name of your project)
3. Choose the directory where the project should be initiated
4. `Create Project`

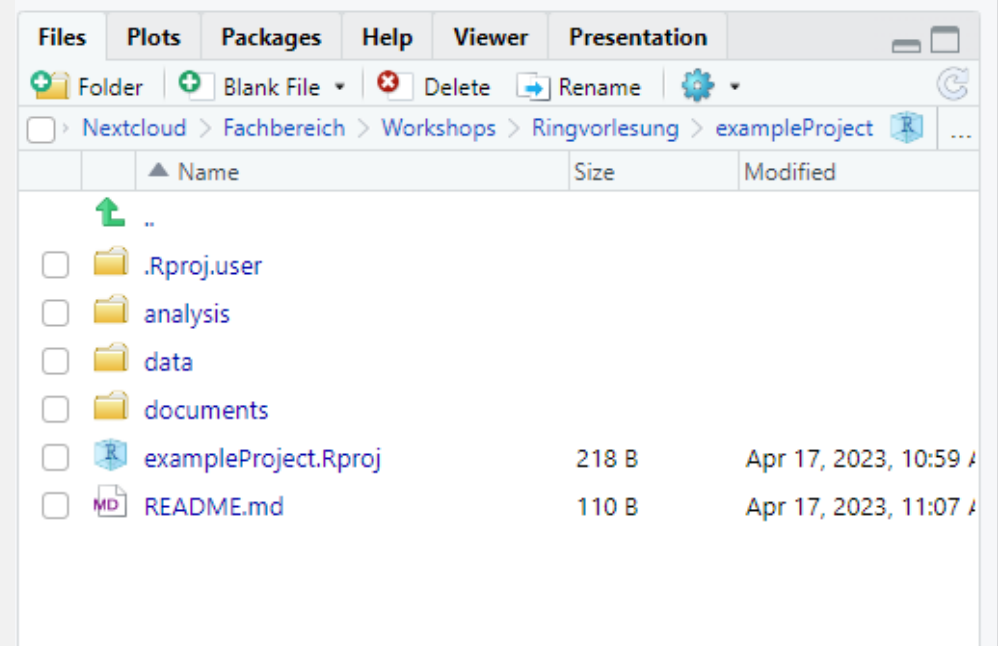
Associate an existing folder with an R Studio Project:

1. `File -> New Project -> Existing Directory`
2. Choose your project folder
3. `Create Project`

Navigate an R Studio Project

You can use the **Files** pane in R Studio to interact with your project folder:

- Navigate and open files
- Create files and folders
- Rename and delete
- ...



Set up your project

R Studio offers a lot of settings and options.

So have a ☕ and check out **Tools -> Global Options** and all the other buttons.

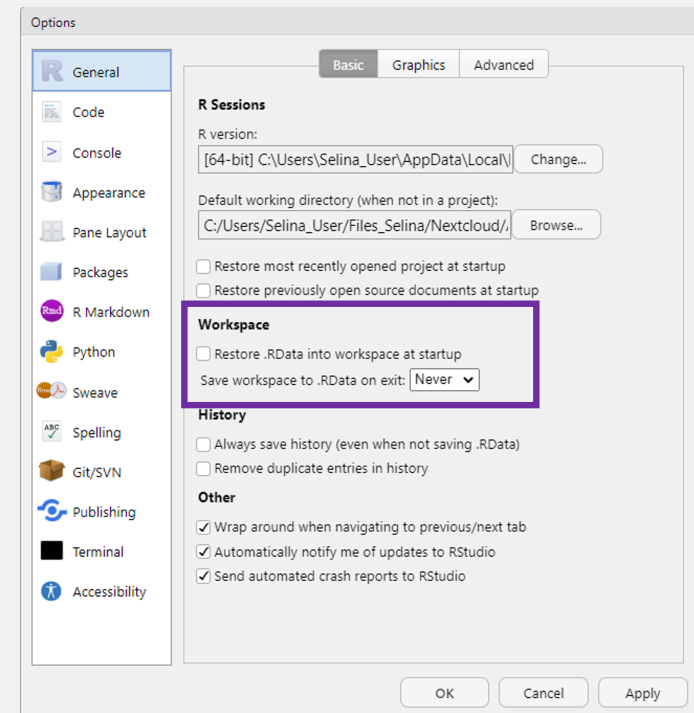
- **R Studio cheat sheet** that explains all the buttons
- Update R Studio from time to time to get new settings (**Help -> Check for Updates**)

Set up your project

R Studio offers a lot of settings and options.

Most important setting for reproducibility:

- Never save or restore your work space as **.Rdata** -> You always want to start working with a clean slate



Name your files properly

Your collaborators and your future self will love you for this.

Principles ¹

File names should be

1. Machine readable
2. Human readable
3. Working with default file ordering

1. Machine readable file names

Names should allow for easy **searching**, **grouping** and **extracting information** from file names.

- No space & special characters

Bad examples ❌

 2023-04-20 temperature göttingen.csv

 2023-04-20 rainfall göttingen.csv

Good examples ✔️

 2023-04-20_temperature_goettingen.csv

 2023-04-20_rainfall_goettingen.csv

2. Human readable file names

Which file names would you like to read at 4 a.m. in the morning?

- File names should reveal the file content
- Use separators to make it readable

Bad examples ❌

 01preparedataforanalysis.R

 01firstscript.R

Good examples ✔️

 01_prepare-data-for-analysis.R

 01_lm-temperature-trend.R

3. Default ordering

If you order your files by name, the ordering should make sense:

- (Almost) always put something numeric first
 - Left-padded numbers (01, 02, ...)
 - Dates in YYYY-MM-DD format

Chronological order

 2023-04-20_temperature_goettingen.csv

 2023-04-21_temperature_goettingen.csv

Logical order

 01_prepare-data.R

 02_lm-temperature-trend.R

Let's start coding

Write beautiful code



Artwork by [Allison Horst](#), CC BY 4.0

Standard code structure

1. General comment with purpose of the script, author, ...
2. `library()` calls on top
3. Set default variables and global options
4. Source additional code
5. Write the actual code, starting with loading all data files

```
# This code replicates figure 2 from the  
# Baldauf et al. 2022 Journal of Ecology paper.  
# Authors: Selina Baldauf, Jane Doe, Jon Doe  
  
library(tidyverse)  
library(vegan)  
  
# set defaults  
input_file <- "data/results.csv"  
  
# source files  
source("R/my_cool_function.R")  
  
# read input  
input_data <- read_csv(input_file)
```

Mark sections

- Use comments to break up your file into sections

```
# Load data -----  
  
input_data <- read_csv(input_file)  
  
# Plot data -----  
  
ggplot(input_data, aes(x = x, y = y)) +  
  geom_point()
```

- Insert a section label with **Ctrl/Cmd + Shift**
- Navigate sections in the file outline

Modularize your Code

- Don't put all your code into one long file (hard to maintain)
 - Write multiple files that can be called sequentially
 - E.g. `01_prepare-data.R`, `02_lm-temperature-trend.R`, `03_plot-temperature-trends.R`
 - Write functions that can be called in other scripts
 - Use the `source()` function to source these files
 - Have one main workflow script that calls these functions sequentially

Use save paths

To read and write files, you need to tell R where to find them.

Common workflow: set **working directory** with `setwd()`, then read files from there. But to this *Jenny Bryan* said:

If the first line of your R script is

```
setwd("C:\\Users\\jenny\\path\\that\\only\\I\\have")
```

I will come into your office and SET YOUR COMPUTER ON FIRE 🔥.

Why?

This is **100% not reproducible**: Your computer at exactly this time is (probably) the only one in the world that has this working directory

...

Avoid `setwd()` if it is possible in any way!

Avoid `setwd()`

Use R Studio projects

- Project root is automatically the working directory
- Give your project to a friend at it will work on their machine as well

Instead of

```
# my unique path from hell with white space and special characters
setwd("C:/Users/Selina's PC/My Projects/Göttingen Temperatures/temperatures")

read_csv("data/2023-04-20_temperature_goettingen.csv")
```

You just need

```
read_csv("data/2023-04-20_temperature_goettingen.csv")
```

If you don't use R Studio Projects, have a look at the [{here}](#) package for reproducible paths

Coding style - Object names

- Variables and function names should only have lowercase letters, numbers, and _
- Use **snake_case** for longer variable names
- Try to use concise but meaningful names

```
# Good
day_one
day_1

# Bad
DayOne
dayone
first_day_of_the_month
dm1
```

Coding style - Spacing

- Always put spaces after a comma

```
# Good
```

```
x[, 1]
```

```
# Bad
```

```
x[,1]
```

```
x[,1]
```

```
x[,1]
```

Coding style - Spacing

- Always put spaces after a comma
- No spaces around parentheses for normal function calls

```
# Good  
mean(x, na.rm = TRUE)
```

```
# Bad  
mean(x, na.rm = TRUE)  
mean(x, na.rm = TRUE)
```

Coding style - Spacing

- Always put spaces after a comma
- No spaces around parentheses for normal function calls
- Spaces around most operators (<-, ==, +, etc.)

```
# Good
height <- (feet * 12) + inches
mean(x, na.rm = TRUE)

# Bad
height <- feet * 12 + inches
mean(x, na.rm = TRUE)
```

Coding style - Spacing

- Always put spaces after a comma
- No spaces around parentheses for normal function calls
- Spaces around most operators (`<-`, `==`, `+`, etc.)
- Spaces before pipes (`%>%`, `|>`) followed by new line

```
# Good
iris %>%
  group_by(Species) %>%
  summarize_if(is.numeric, mean) %>%
  ungroup()

# Bad
iris %>%
  group_by(Species) %>%
  summarize_all(mean) %>%
  ungroup()
```


Coding style - Spacing

- Always put spaces after a comma
- No spaces around parentheses for normal function calls
- Spaces around most operators (`<-`, `==`, `+`, etc.)
- Spaces before pipes (`%>%`, `|>`) followed by new line
- Spaces before `+` in ggplot followed by new line

```
# Good
ggplot(aes(x = Sepal.Width, y = Sepal.Length, color = Species)) +
  geom_point()

# Bad
ggplot(aes(x = Sepal.Width, y = Sepal.Length, color = Species)) +
  geom_point()
```

Coding style - Line width

Try to limit your line width to 80 characters.

- You don't want to scroll to the right to read all code
- 80 characters can be displayed on most displays and programs
- Split your code into multiple lines if it is too long
 - See this grey vertical line in R Studio?

```
# Bad
iris %>%
  group_by(Species) %>%
  summarise(Sepal.Length = mean(Sepal.Length), Sepal.Width = mean(Sepal.Width), Species = n_distinct(Speci

# Good
iris %>%
  group_by(Species) %>%
  summarise(
    Sepal.Length = mean(Sepal.Length),
    Sepal.Width = mean(Sepal.Width),
    Species = n_distinct(Species)
  )
```

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Coding style

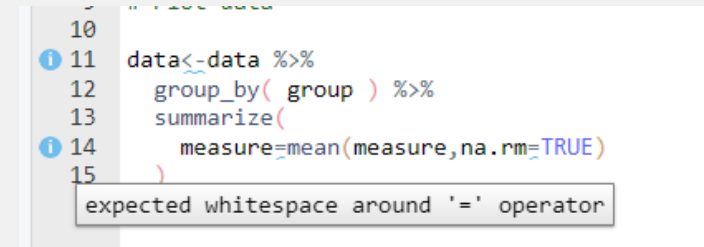
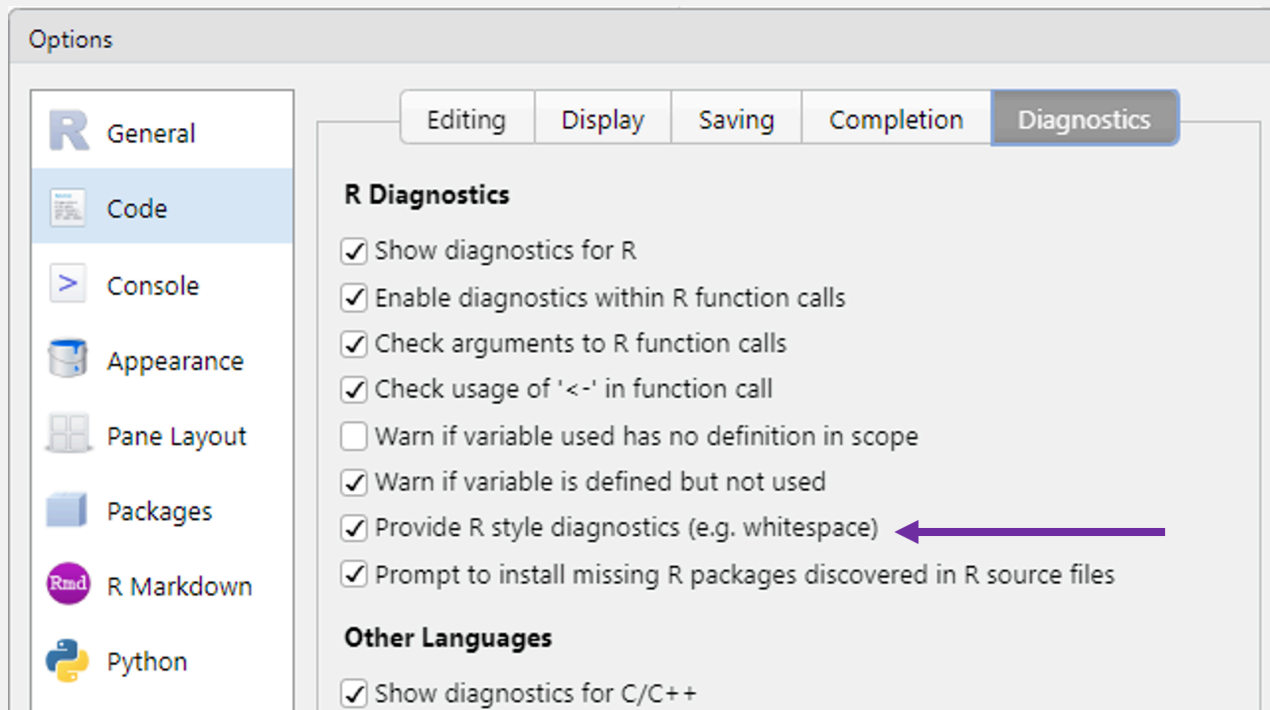
Do I really have to remember all of this?

Luckily, no! R and R Studio provide some nice helpers

Coding style helpers - R Studio

R Studio has style diagnostics that tell you where something is wrong

- Tools -> Global Options -> Code -> Diagnostics

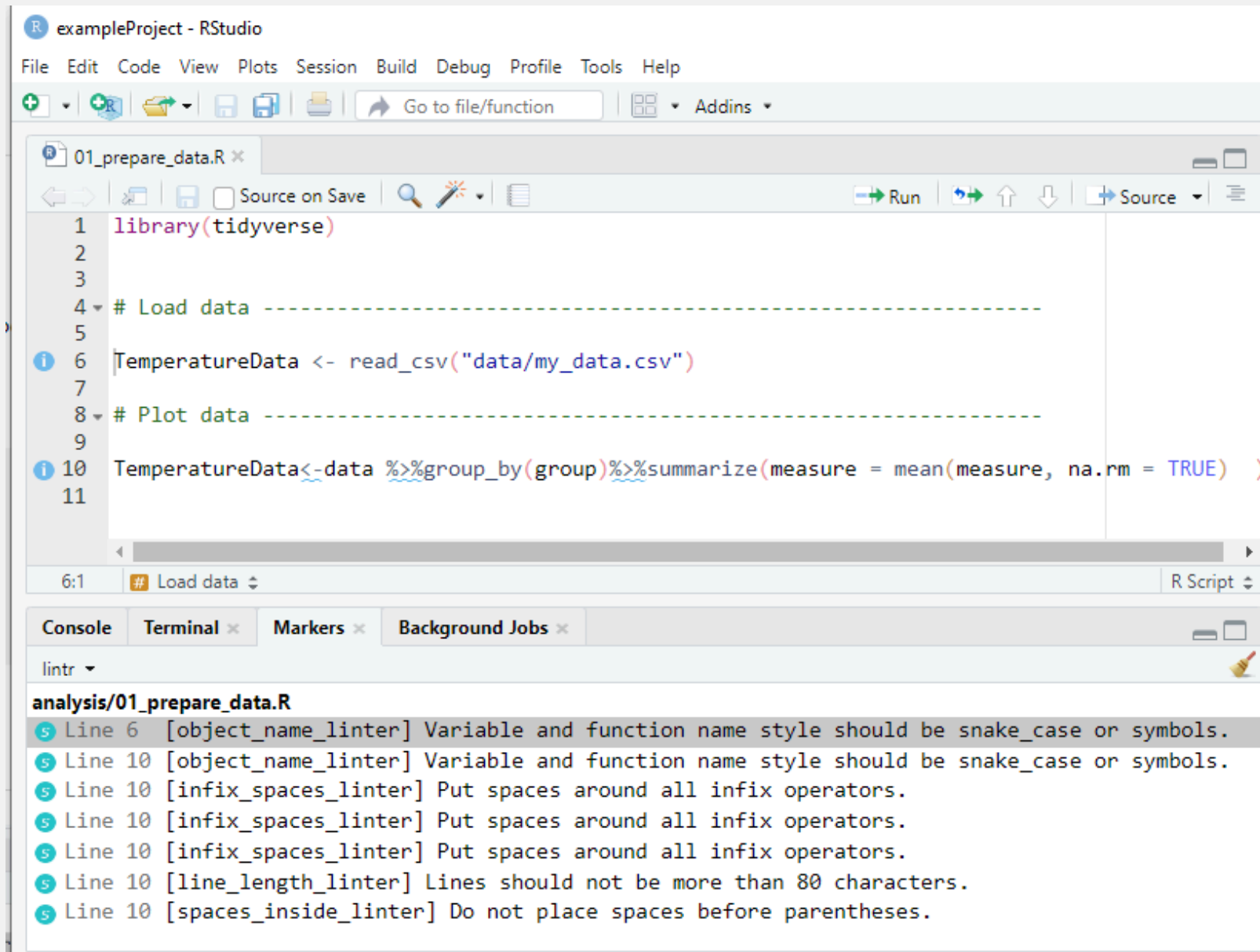


Coding style helpers - {lintr}

The `lintr` package analyses your code files or entire project and tells you what to fix.

```
# install the package before you can use it
install.packages("lintr")
# lint specific file
lintr::lint(filename = "analysis/01_prepare_data.R")
# lint a directory (by default the whole project)
lintr::lint_dir()
```

Coding style helpers - {lintr}



The screenshot shows the RStudio interface with a project named 'exampleProject'. The source editor displays the file '01_prepare_data.R' with the following R code:

```
1 library(tidyverse)
2
3
4 # Load data -----
5
6 TemperatureData <- read_csv("data/my_data.csv")
7
8 # Plot data -----
9
10 TemperatureData <- data %>% group_by(group) %>% summarize(measure = mean(measure, na.rm = TRUE))
11
```

The console at the bottom shows the output of the lintr package, listing several linting errors for the file 'analysis/01_prepare_data.R':

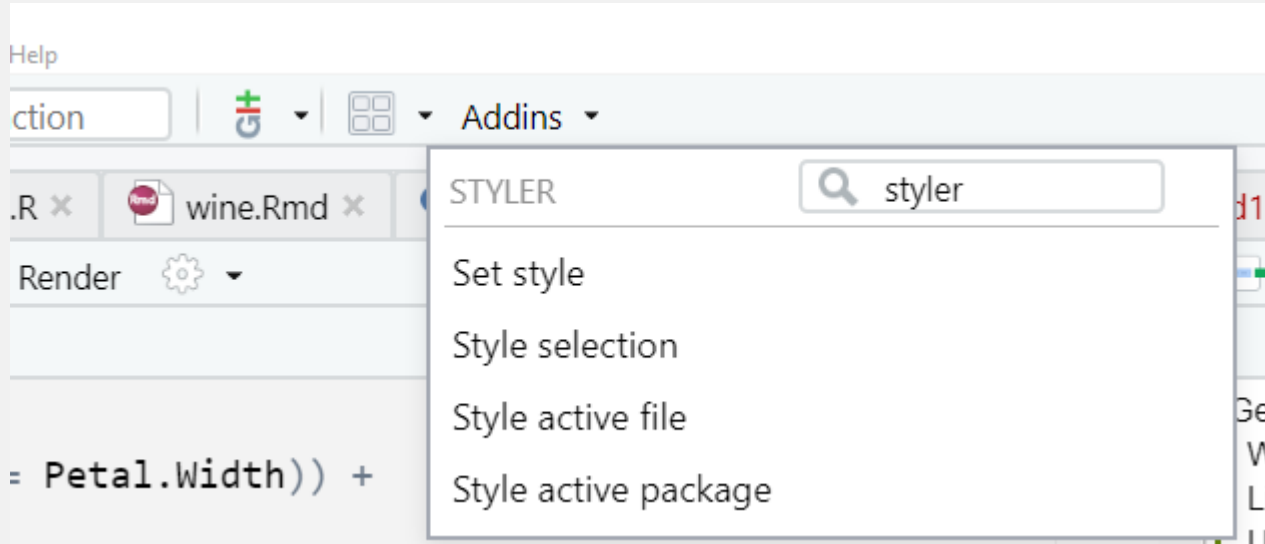
- Line 6 [object_name_linter] Variable and function name style should be snake_case or symbols.
- Line 10 [object_name_linter] Variable and function name style should be snake_case or symbols.
- Line 10 [infix_spaces_linter] Put spaces around all infix operators.
- Line 10 [infix_spaces_linter] Put spaces around all infix operators.
- Line 10 [infix_spaces_linter] Put spaces around all infix operators.
- Line 10 [line_length_linter] Lines should not be more than 80 characters.
- Line 10 [spaces_inside_linter] Do not place spaces before parentheses.

Coding style helpers - {styler}

The [styler package](#) automatically styles your files and projects according to the tidyverse style guide.

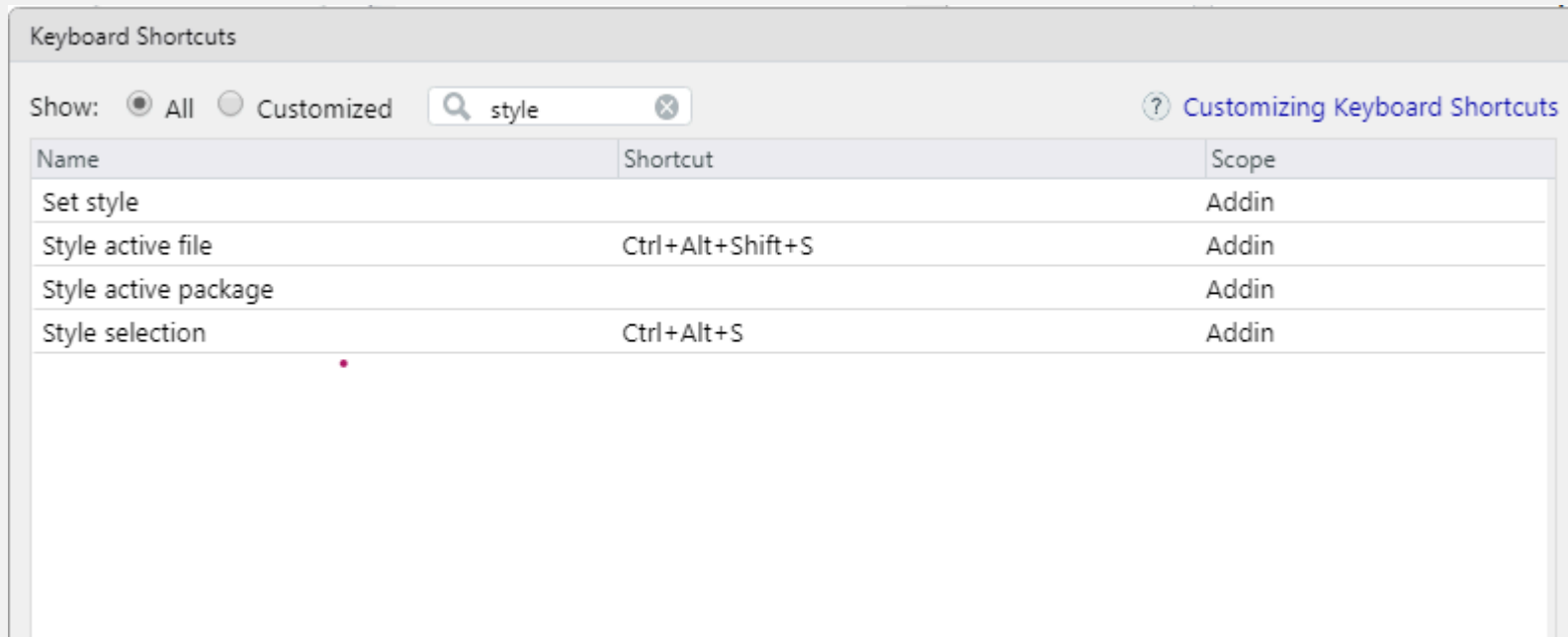
```
# install from CRAN  
install.packages("styler")
```

- Use the R Studio [Addins](#) for styler:



Coding style helpers - {styler}

- Pro-Tip: Add a custom keyboard short cut to style your files
 - Tools -> Modify Keyboard Shortcuts



Manage dependencies with {renv}

Idea: Have a **project-local environment** with all packages needed by the project

- Keep log of the packages and versions you use
- Restore the local project library on other machines



Why this is useful?

- Code will still work even if packages upgrade
- Collaborators can recreate your local project library with one function
- Explicit dependency file states all dependencies

Check out the [renv website](https://rstudio.github.io/renv/) for more information

Manage dependencies with {renv}

```
# Get started  
install.packages("renv")
```

Very simple to use and integrate into your project workflow:

```
# Step 1: initialize a project level R library  
renv::init()  
# Step 2: save the current status of your library to a lock file  
renv::snapshot()  
# Step 3: restore state of your project from renv.lock  
renv::restore()
```

- Your collaborators only need to install the `renv` package, then they can also call `renv::restore()`
- When you create an R Studio project there is a check mark to initialize with `renv`

Summary

Clean projects and workflows ...

... allow you and others to work productively.



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Take aways

There are a lot of things that require minimal effort and that you can start to implement into your workflow NOW

1. Use R Studio projects -> Avoid `setwd()`!
2. Keep your projects clean
 - Sort your files into folders
 - Give your files meaningful names
3. Use `styler` to style your code automatically
4. Use `lintr` and let R analyse your project
5. Consider `renv` for project local environments

Outlook

Of course there is much more:

- Version control with Git
- Using R packages to build a research compendium
- Docker containers for full reproducibility
- ...

But this is for another time

Next lecture

Write reproducible documents with Quarto

Quarto (the successor of rmarkdown) is a powerful tool that enables the seamless integration of code (R, Python, and more) and its output into a variety of formats such as reports, research papers, presentations, and more. This tool streamlines the process of creating reproducible workflows by eliminating the need to copy and paste figures, tables, or numbers. During this lecture, you'll learn the fundamentals of Quarto and explore practical use cases that you can implement in your data analysis workflow.

. . .



11th May



4-5 p.m.



Webex

Thank you for your attention
:)

Questions?

References

- [What they forgot to teach you about R book](#) by Jenny Bryan and Jim Hester
- [Blogpost](#) by Jenny Bryan on good project-oriented workflows
- [R best practice blogpost](#) by Krista L. DeStasio
- [Book about coding style for R: The tidyverse style guide](#)
- [The Turing way book](#) General concepts and things to think about regarding reproducible research
- [renv package website](#)
- [styler package website](#)
- [lintr package website](#)